

Hao Cao 曹浩

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RESEARCH INTERESTS

Planetary geophysics and space environment · planetary magnetism and dynamos · gravitational and electromagnetic sounding of planetary interiors · planetary exploration, including space missions and space-based magnetometry.

HONORS AND AWARDS

2025	Sloan Research Fellowship in Physics , Alfred P. Sloan Foundation
Mar 2022	Certificate of Distinction in Teaching , Harvard University
Aug 2018	NASA Group Achievement Award , Cassini MAG Team, NASA
Sep 2012	Eugene B. Waggoner Scholarship , UCLA
May 2012	NASA Earth and Space Science Fellowship , NASA

PARTICIPATION IN SPACE MISSIONS

2024 – present	Deputy Instrument Lead / Co-I , NASA TRACERS MAG Team
2016 – present	Co-Investigator , ESA JUICE J-MAG Team
2014 – present	Member , Juno Science Team
2015 – 2017	Cassini Participating Scientist , Cassini Mission
2014 – 2017	Member , Cassini MAG Team

SCIENTIFIC INSTRUMENT DEVELOPMENT

2024 – present	Principal Investigator , NASA DALI Project L-MAG (Lunar Magnetometer)
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ACADEMIC APPOINTMENTS

2026 – present	Associate Professor , University of California, Los Angeles
2022 – 2026	Assistant Professor , University of California, Los Angeles
2017 – 2022	Research Associate , Harvard University
2014 – 2017	Postdoctoral Scholar , California Institute of Technology

EDUCATION

Jun 2014	Ph.D. , Geophysics and Space Physics, UCLA
Jun 2011	M.S. , Geophysics and Space Physics, UCLA
Jun 2009	B.S. , Geophysics and Space Physics, University of Science and Technology of China

VISITING POSITIONS

Sep – Dec 2022	Simons Fellow , Isaac Newton Institute for Mathematical Sciences, University of Cambridge
2022 – 2024	Associate , Harvard University
2018 – present	Academic Visitor in Physics , Imperial College London
2017 – 2020	Visitor in Planetary Science , California Institute of Technology

SELECTED PUBLICATIONS

1. Agiwal, O., **Cao, H.**, Hsu, H.W., et al. (2024). Current events at Saturn: ring–planet electromagnetic coupling. *Planetary Science Journal* 5, 134. doi:10.3847/PSJ/ad4343
2. Bloxham, J., **Cao, H.**, Stevenson, D.J., et al. (2024). A rapidly time-varying equatorial jet in Jupiter’s deep interior. *Nature* 627, 64–66. doi:10.1038/s41586-024-07046-3
3. **Cao, H.**, Bloxham, J., Park, R.S., et al. (2023). Strong resemblance between surface and deep zonal winds inside Jupiter revealed by high-degree gravity moments. *The Astrophysical Journal* 959, 78. doi:10.3847/1538-4357/ad0cbb
4. **Cao, H.**, Dougherty, M.K., Hunt, G.J., et al. (2023). Saturn’s magnetic field at unprecedented detail achieved by Cassini’s close encounters. In *Saturn after the Cassini Grand Finale*, Cambridge University Press. arXiv:2301.02756
5. Yadav, R.K., **Cao, H.**, Bloxham, J. (2022). A global simulation of the dynamo, zonal jets, and vortices on Saturn. *The Astrophysical Journal* 940, 185. doi:10.3847/1538-4357/ac9d94
6. Agiwal, O., **Cao, H.**, Cowley, S.W.H., et al. (2021). Constraining the temporal variability of neutral winds in Saturn’s low-latitude ionosphere using magnetic field measurements. *JGR: Planets* 126, e2020JE006578. doi:10.1029/2020JE006578
7. **Cao, H.**, Dougherty, M.K., Hunt, G.J., et al. (2020). The landscape of Saturn’s internal magnetic field from the Cassini Grand Finale. *Icarus* 344, 113541. doi:10.1016/j.icarus.2019.113541

Full publication list: sites.epss.ucla.edu/hcao/publications

NATIONAL AND INTERNATIONAL COMMITTEES

2025 – present	Member , JUICE–Clipper Steering Committee, ESA JUICE Project and NASA Europa Clipper Project
2023 – 2026	UCLA Representative to the Council of Institutions, Universities Space Research Association

PROFESSIONAL SERVICE

Conference roles. SOC Co-chair, NASA TRACERS Science Team Meeting (2024); LOC Co-chair, Russell Symposium (2024); SOC Member, DPS55/EPSC, San Antonio (2023); Session Convener or Chair, AGU Fall Meeting (2018, 2023–2025), DPS55/EPSC (2023), DPS48/EPSC11, Pasadena (2016).

Journal referee. Nature; Nature Astronomy; Nature Geoscience; Nature Communications; PNAS; Space Science Reviews; The Astrophysical Journal; Icarus; JGR Planets; Geophysical Research Letters; Earth and Planetary Science Letters; Progress in Earth and Planetary Science; MNRAS; Scientific Reports; Advances in Space Research; Atmosphere; Earth and Planetary Physics; Frontiers in Physics.

INVITED SEMINARS AND COLLOQUIA

Caltech, Planetary Science Seminar (Apr 2026) · Imperial College London, Physics Department Seminar (Nov 2025) · UC Berkeley, CIPS Seminar (Apr 2025) · Caltech, Planetary Science Seminar (Feb 2023) · JPL, ICE Seminar (Jan 2023) · University of Washington, ESS Colloquium (Apr 2022) · UCLA, Space Physics Seminar (Feb 2022) · ETH Zürich, Earth Sciences ERC Seminar (Jan 2021) · UC Berkeley, Astronomy/EPS Joint Colloquium (Nov 2019) · Boston University, Center for Space Physics Seminar (Mar 2019) · Georgia Tech, EAS Seminar (Feb 2019) · Imperial College London, Space & Atmospheric Physics Seminar (Mar 2018) ·

University of Leicester, Astro Seminar (Mar 2018) · University of Cambridge, DAMTP Astrophysics Seminar (Feb 2018) · MIT, EAPS Planetary Lunch (Feb 2018) · Cornell, Astronomy and Space Sciences Special Seminar and Colloquium (Feb 2018) · Harvard, EPS GSPD Seminar (Feb 2018) · Carnegie DTM, Astro Seminar (Jan 2018) · Imperial College London, Space & Atmospheric Physics Seminar (Nov 2017) · UT Austin, Institute of Geophysics Seminar (Nov 2016) · JPL, Planetary Science Seminar (Sep 2016) · UC Santa Cruz, Whole Earth Seminar (Feb 2016) · UCLA, Planet Lunch (Jan 2016) · High Altitude Observatory, Boulder, Colloquium (Apr 2015) · Caltech, Yuk Lunch Seminar (Mar 2015) · UCLA, iPLEX Lunch Seminar (May 2013).